

THE BN-350 AND BOR FAST REACTORS

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THE BN-350 REACTOR

In accordance with the plan for the construction of atomic stations, the BN-350 fast reactor station is being built on the shores of the Caspian Sea in the region of the Mangyshlak peninsula in the U.S.S.R. Designed to produce 150,000 kW of electric power and 12,000 tons of fresh water per day, this is the first large fast power reactor in the U.S.S.R. For the initial operation period the parameters chosen are ones which have already been achieved and experimentally justified. The reactor can operate, without core modification, under various conditions, using both U-235 and plutonium as fuel.

The main technical details and structural features of the BN-350 reactor were given at the 3rd Geneva Conference on the peaceful uses of atomic energy and in a paper at the Detroit Conference in 1965. At present the experimental work on the most important components of the reactor is being completed and the manufacture of systems and equipment is under way. The following contains a description and test results on the individual systems and operating conditions of the BN-350 reactor.

1. Reactor Refuelling System

The refuelling system includes two rotating plugs, the refuelling mechanism, two elevators, a refuelling box with transfer mechanism, and two transfer drums (one for new, the other for spent assemblies). The assembly refuelling mechanism is located in the small rotating plug, which in turn is located eccentrically in the large rotating plug. Rotation of the two plugs brings the refuelling mechanism over any socket of the core and also the elevator socket. The elevators are hoists, in the carriages of which there are sockets for holding the assemblies. Movement of the carriage along an

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inclined guide transfers the assemblies by the elevators from the refuelling mechanism to the transfer mechanism and vice versa.

At the point of contact of the elevator with the transfer mechanism, the top of the assembly above the sodium level. This avoids the gripping device of the transfer mechanism having to dip into the sodium. The assembly transfer mechanism is located in the refuelling box. Movement of the mechanism along the box guides the gripping device to the elevator sockets and the transfer drum sockets. The latter grips the assembly, raises it into the box and transports it to the mouth of the spent assembly drum. After this it lowers the assembly into one of the sockets of the drum.

On the opposite journey the transfer mechanism picks up the new assembly from the other drum and transfers it to the elevator which is prepared for the lowering of the new assemblies. The elevator lowers the assembly into the lower position, in which it is gripped by the refuelling mechanism and is conveyed to the socket intended for the new assembly.

The assembly refuelling cycle is repeated until all the spent assemblies are discharged and the necessary quantity of new assemblies are charged into the reactor. The refuelling cycle (about forty assemblies) lasts about 3 days, including preparation for refuelling and start-up of the reactor after refuelling.

Refuelling Equipment

The refuelling equipment consists of the rotating plugs (large and small) and the refuelling mechanism. The rotating plugs guide the refuelling mechanism to the assemblies and are a part of the top biological shield of the reactor.

Rotation of the plugs is on ball bearings consisting of ball races and balls. The reactor is sealed (in the gaps between the plugs) with a liquid Sn-Bi eutectic with a melting point of 138°C. When the plugs are rotating the alloy is melted by electric heaters fitted in the bath with the alloy.

During operation of the reactor the alloy is in a solidified condition, giving the necessary leak-tightness to the reactor. Apart from the freeze seal there is a duplicate mechanical seal consisting of a rubber ring bolted against the sealing faces. For the refuelling period the rubber ring is slightly raised, freeing the plugs for rotation. The top parts of the plugs are cooled by air blown through special channels.

Refuelling Mechanism

The design of the refuelling mechanism is shown in Fig. 1. The mechanism consists of the gripping device with rods, a guide tube with support flange, a lifting device, a rotating device, and drives.

All the actuating elements are fitted in the housing of the mechanism. The gripping device engages with the top of the assembly, removes the assembly

from the socket, holds it during transfer and places it into the drum socket. Control of the whole complex of mechanisms of the refuelling system and supervision of the operation is effected from the control desk in accordance with a programme defining the sequence of operation of the mechanisms.

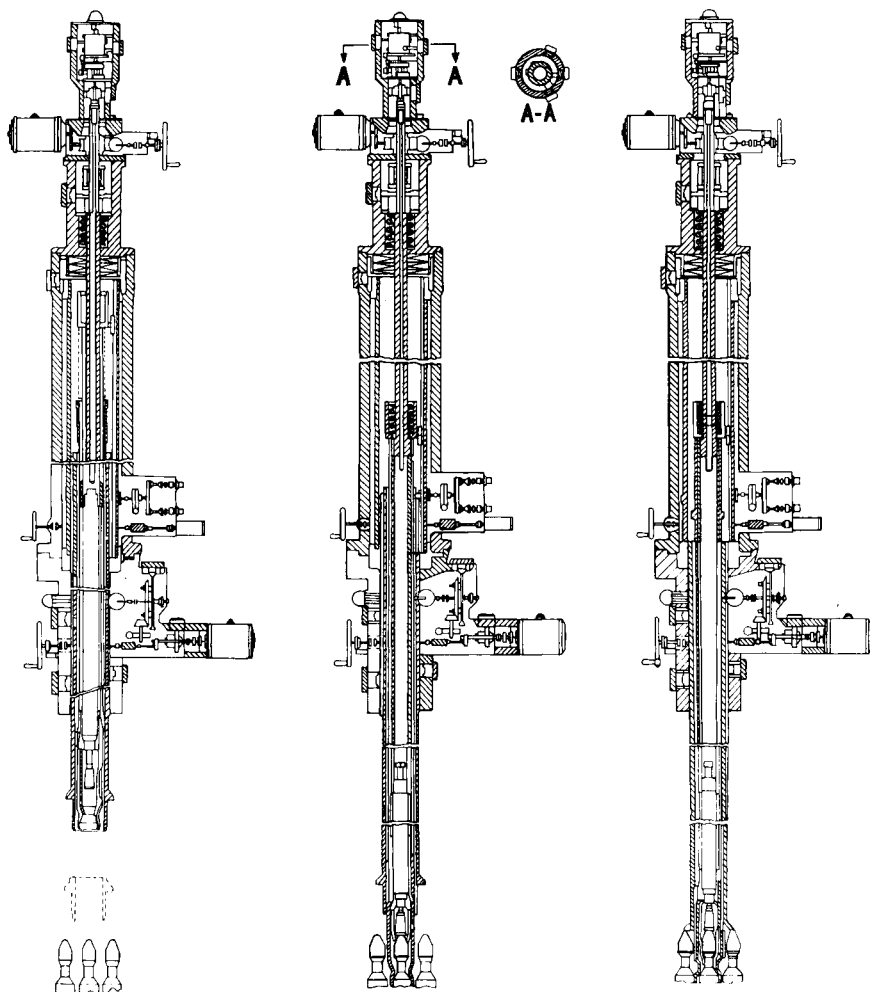


FIG. 1. Refuelling mechanism.

The design of the refuelling mechanism provides for refuelling of the core and blanket assemblies and the rods of the safety and control system.

In the refuelling and transfer mechanisms there are special devices to control the gripping and releasing of the assemblies. The spent assemblies are transported from the reactor to the transfer drum in an argon atmosphere.

Where necessary, in transfer of the assemblies, forced cooling of the assembly is possible by circulation of argon, using a special blower fitted on the assembly transfer mechanism.

Refuelling takes place in the following sequence: the assembly to be removed is gripped by the refuelling mechanism and is transferred to the elevator. The elevator raises it to the point of contact with the transfer mechanism. The guide tube serves as a support to hold down the adjacent assemblies and the guide for the gripping device. It prevents the grab accidentally opening during lifting and lowering of the gripping device with the assembly. The assembly can be rotated through 180° .

The hoist is for lifting and lowering the grab and rod and consists of a screw and ball nut. The drives are fitted with electric motors and manual control and there is a signalling device which indicates whether the gripping and release of the assembly has been carried out.

All the elements of the refuelling equipment, the refuelling mechanisms, and these mechanisms as a whole have undergone appropriate tests. Leak tests have been carried out on the seal with liquid and solid Sn-Bi alloy. A special rig was constructed for the tests, simulating the design of the seal of the rotating plugs.

The purpose of the work was to select structural materials compatible with the Sn-Bi alloy and to ensure a reliable seal. As a result of the tests a material was chosen for the parts of the seal unit in direct contact with the alloy (steel 20) and the type of coating was determined to ensure reliable wetting.

The tests were carried out at an alloy temperature up to 200°C and argon pressure of 1.5 atm.g. The duration of the tests with rotation of the plugs corresponds to about 10 years' operation of the rotating plugs in refuelling. During the experiment no leakage of argon through the seal was detected. Helium tests on the seal also confirmed there was no gas leakage. To prevent oxidation of the Sn-Bi alloy a silicone vacuum oil is used which meets the prescribed requirements of stability and compatibility with the Sn-Bi alloy and structural materials of the seal. The thickness of the layer of protective liquid is about 80 mm.

The purpose of the tests on the refuelling mechanism was to determine the life of the mechanism under operating conditions. The tests were carried out on a sodium rig with sodium circulating at a temperature of 500°C . To simulate the core of the reactor on the rig, a group of assemblies was used, charging of which was carried out with the mechanism. The assembly was engaged by the mechanism, raised to a height of about 3600 mm, returned to the socket and the mechanism disengaged. In all the mechanism operated for more than 6000 hours.

The tests were carried out both with the assembly and the grab of the mechanism in coaxial positions, and with eccentricity of up to 20 mm. In all cases the grab reliably engaged with the assembly. The most suitable material

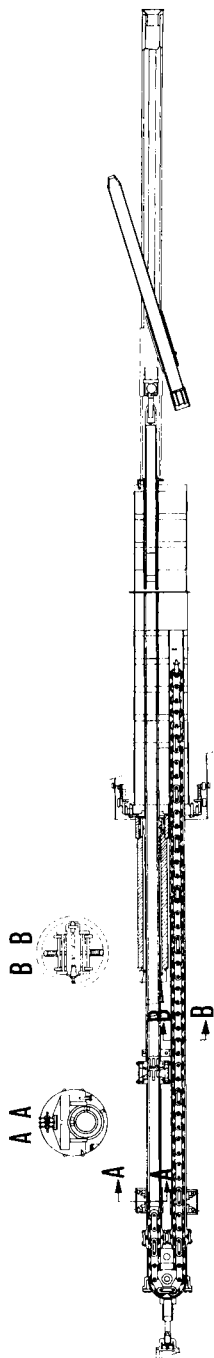


FIG. 2. Charge/discharge hoist.

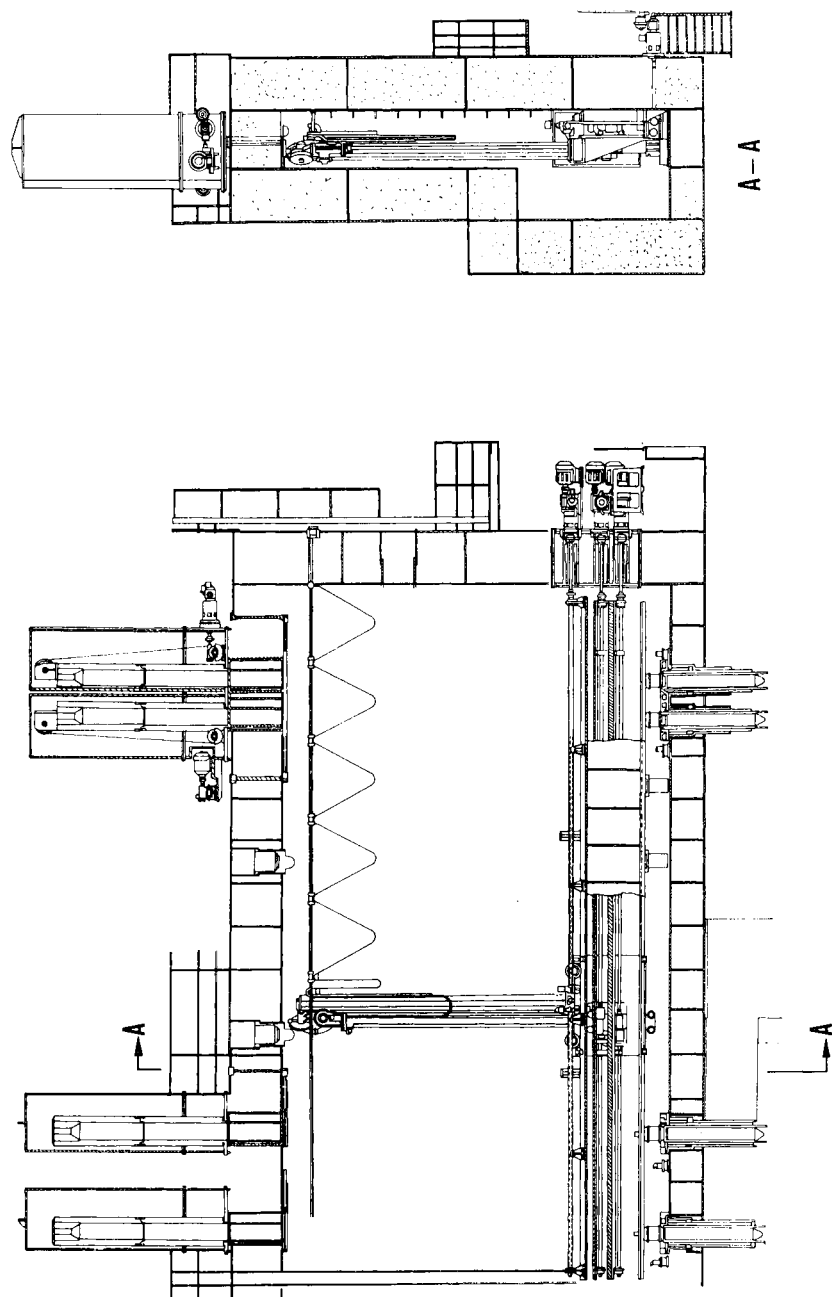


FIG. 3. Assembly transfer mechanism.

for the grab and friction pairs operating in sodium was found to be a diffusion chromized stainless steel type 18Cr9Ni. Figure 2 shows the design of the elevator, which is a lifting mechanism, the carriage of which moves along a special guide at an angle of 17° from the vertical. In the carriage there is a fixed sleeve in which the transported assemblies are placed. The carriage is moved both by motor and by hand. The carriage drive consists of a motor, reduction gear, chain transmission, and rod. Table 1 contains technical data on the elevator.

TABLE 1

1. Travel of carriage	4950 mm
2. Rate of movement of carriage	3.14 m/min
3. Slowed down rate of travel of carriage	0.7 m/min
4. Carriage lifting (lowering) time	2 min
5. Motor rating	0.83 kW

The assembly transfer mechanism (ATM) (Fig. 3) consists of a rack with grab, a rack lift reduction gear, a reduction gear frame, and a girder with guides. The rack is made in the form of a tube, with teeth cut along the generating line. The frame with the reduction gear and rack moves horizontally along the box on guides. The rack and grab chain are driven by feed shafts and movement of the ATM is effected by a lead screw. The technical data of the ATM are given in Table 2:

TABLE 2

1. Rate of horizontal movement of ATM	7 m/min
2. Rate of vertical movement of rack and reduction gear	3.5 m/min
3. Rate of movement of grab	3.5 m/min
4. Rate of rise of plugs	3.45 m/min

In 1966 tests will be carried out on an actual example of the elevator. For these tests a sodium rig has been built with sodium circulating at 500°C . On the rig the performance of the elevator components will be checked and problems of the installation, replacement, and repair of the mechanism will be dealt with.

During 1966-7 it is intended, on a specially built rig at the manufacturer's works, to carry out a test assembly of the reactor from the actual parts and components being manufactured. On the rig the core and blanket will be set up with the pressure header, rotating plugs with central column, safety and control system mechanisms, and refuelling system mechanisms. The core and blanket will be made up of simulated assemblies, which will be used later in start-up of the reactor.

The purpose of the tests on the test assembly rig is:

- (a) to check the assembly of the reactor zone, components, and mechanisms on the metal structure of the rig simulating the reactor vessel

and transfer box, in accordance with the requirements of the assembly/ installation drawings of the mechanisms;

- (b) to check the operation of each mechanism of the refuelling system separately;
- (c) to check and adjust the electrical equipment of the system for guiding the refuelling mechanism to the top of the assemblies, the electrical control system of the refuelling mechanisms and the interlock system, with a check on the control desk and control panel equipment;
- (d) to check the operation of the mechanisms of the safety and control system and their interaction with the other reactor components;
- (e) to check the operation of the refuelling mechanisms as a whole and carry out a number of experimental refuellings of different parts of the reactor zone for the purpose of developing the technology of assembly loading and unloading;
- (f) to check and develop the technology of replacement of safety and control rods;
- (g) to check and adjust the regular apparatus for installation and repair of the reactor equipment.

Tests on the assembly transfer mechanism will be carried out on the test assembly rig.

2. Safety and Control System

The safety, regulation, and control system of the BN-350 reactor is part of the overall centralized control system of the plant which consists of the following parts:

- (a) system of measurement and control of the reactor power and period (including subcritical conditions);
- (b) automatic control system;
- (c) reactivity change compensating system; and
- (d) reactor safety system.

The main technical data and a description of the design of the safety and control mechanisms are given in Table 3.

The operating elements of the safety and control system mechanisms of the reactor are absorber rods and compensating assemblies. Each mechanism unit consists of a support with rod, rack mechanism, servo-drive, absorber rod or compensating assembly.

The support and rod is an individual, complete assembly fastened to the flange of the central column of the reactor. The rack mechanism is fastened to the corresponding support with studs, and the joint is sealed by the use of a vacuum rubber gasket.

The servo-drives of the mechanisms are fitted on a special slab. Coupling of the servo-drives to the corresponding rack mechanism is achieved by means of sealed drives.

TABLE 3. MAIN TECHNICAL DATA OF THE SAFETY AND CONTROL SYSTEM ACTUATING MECHANISMS

<i>Mechanism for automatic control</i>	
1. Number of mechanisms	2
2. Number of rods per mechanism	1
3. Radius of movement (from core centre)	98 mm
4. Effectiveness of one rod	0.20%
5. Travel of rod	750 mm
6. Maximum rate of movement	150 mm/sec
<i>Compensator mechanism</i>	
1. Number of independent mechanisms	7
2. Number of assemblies per mechanism	1
3. Effectiveness (total)	2.1%
4. Weight of uranium in assembly	18 kg
5. Travel of assembly	1060 mm
6. Rate of movement	10 mm/sec
7. Pitch of movement	20 mm
<i>Safety mechanism</i>	
1. Number of independent systems	3
2. Effectiveness of system (total)	3.5%
3. Travel of rod	1260 mm
4. Operating time	0.7 s
5. Rate of movement (rising)	5 mm/sec

The Automatic Control Mechanism (Fig. 4)

The mechanism consists of :

- (a) electric motor, rating 0.37 kW;
- (b) worm-type reduction gear;
- (c) rods with grabs;
- (d) induction devices for indicating the position of the rods and grabs;
- (e) drum with cable and counter-weight.

The rotation from the enclosed electric motor is converted via the worm mechanism into translational movement of the rod, on the bottom end of which there is an end fitting with grabs, by means of which it engages with the absorber rod. Control of the grabs is manual, by means of the levels at the top of the mechanism. Here there is provision for checking the engagement and release of the rod and absorber; this is done with a shaft passing inside the rod.

The design of the absorber of the automatic control mechanism is shown in Fig. 5. The absorber includes seven elements each 9.6 mm in diameter, containing boron carbide up to 80% enriched with B-10. To collect the

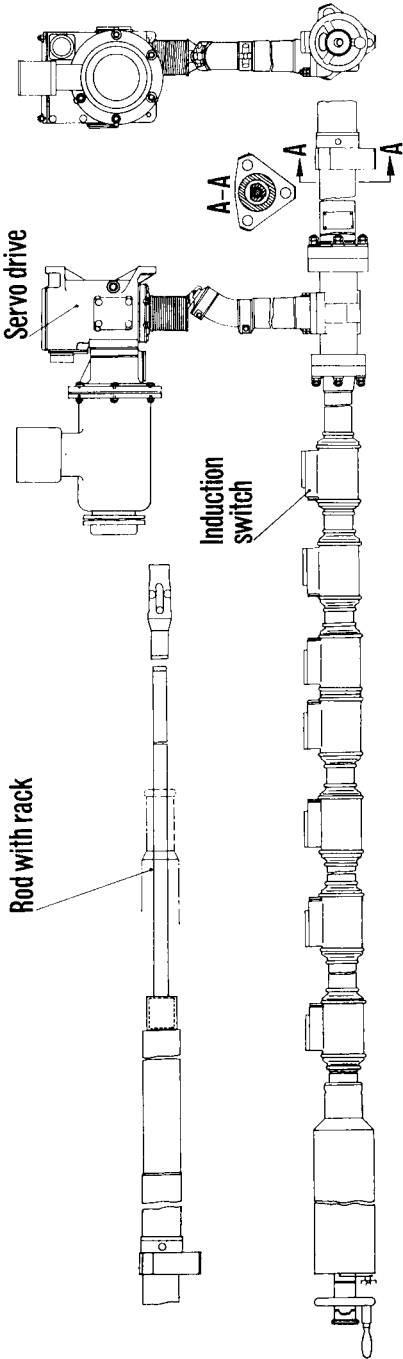


FIG. 4. Automatic control rod actuating mechanism.

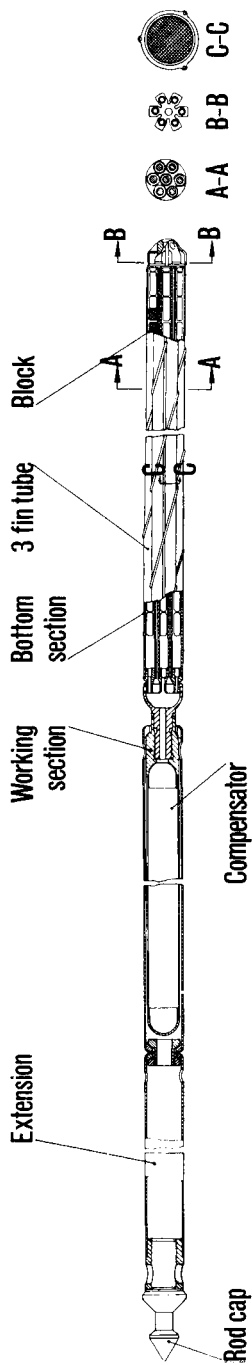


FIG. 5. Automatic control rod.

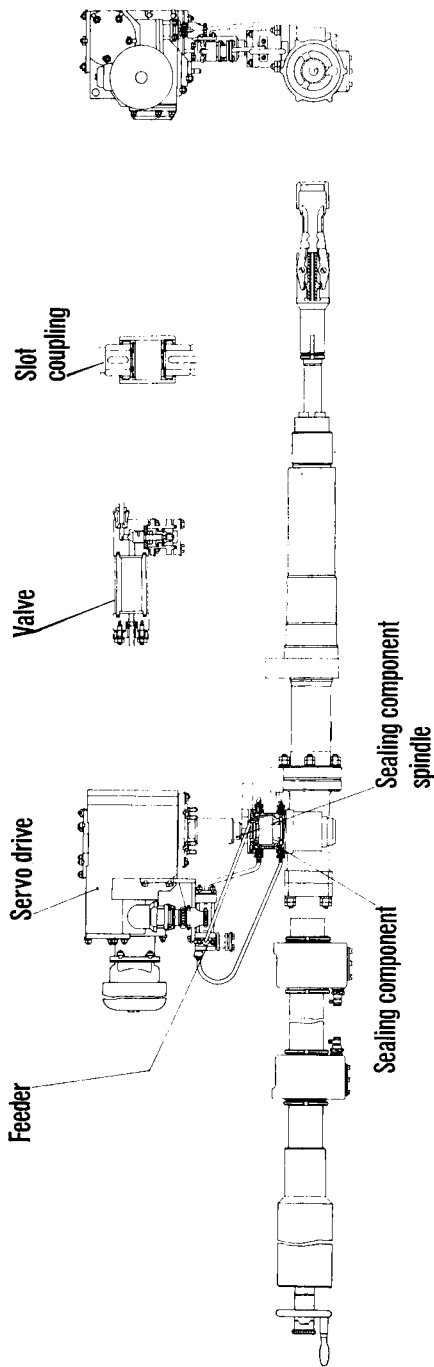


FIG. 6. Operating mechanism of compensator rod.

helium forming as a result of the burnup of the boron, a gas cavity is provided at the top of the absorber.

The Compensator Mechanism (Fig. 6)

The servo-drive of the compensator mechanism is similar to the automatic control servo-drive. Compensation of changes in reactivity is achieved by axial movement of assemblies with active material (burnup compensators) in the core and one assembly with a boron absorber (temperature compensator). A drawing of the burnup compensator is shown in Fig. 7. The compensator contains an active part consisting of fuel elements similar in design to the core elements. The temperature compensator is made of elements containing boron carbide.

The Safety Mechanism (Fig. 8)

The design of the mechanism differs from the automatic control and compensator mechanism by the presence of an accelerating spring ensuring the required operating time and a damper spring for softening the impact force of the moving parts.

The design of the core rod is shown in Fig. 9. The rod includes assemblies consisting of seven boron carbide elements. Tests on the safety and control mechanism have been carried out on the sodium rig at a circulating sodium temperature of 500°C and oxide content of $5 \times 10^{-3}\%$.

The total period of operation of the mechanisms in the sodium was more than 8000 hours. During this period the automatic control mechanism completed about 20,000 double movements, the safety mechanism completed 1600 drops and 400 double movements under slow operation conditions, and the compensating mechanism completed 2500 double movements. During the whole period of the life tests all the mechanisms operated correctly and there were no cases of breakdown or failure to operate.

3. Emergency Shutdown Cooling and Transient Conditions in the BN-350 Plant

Great attention has been paid to the problems of the removal of the after-heat from the reactor when the safety system has operated. The suitability of each system of shutdown cooling was assessed both from the point of view of preventing overheating of the fuel elements, and of preventing the development of dangerous non-steady state thermal stresses in the structural elements of the reactor.

On the basis of these calculations, an emergency shutdown cooling system was designed for the plant. The method makes it possible to calculate the

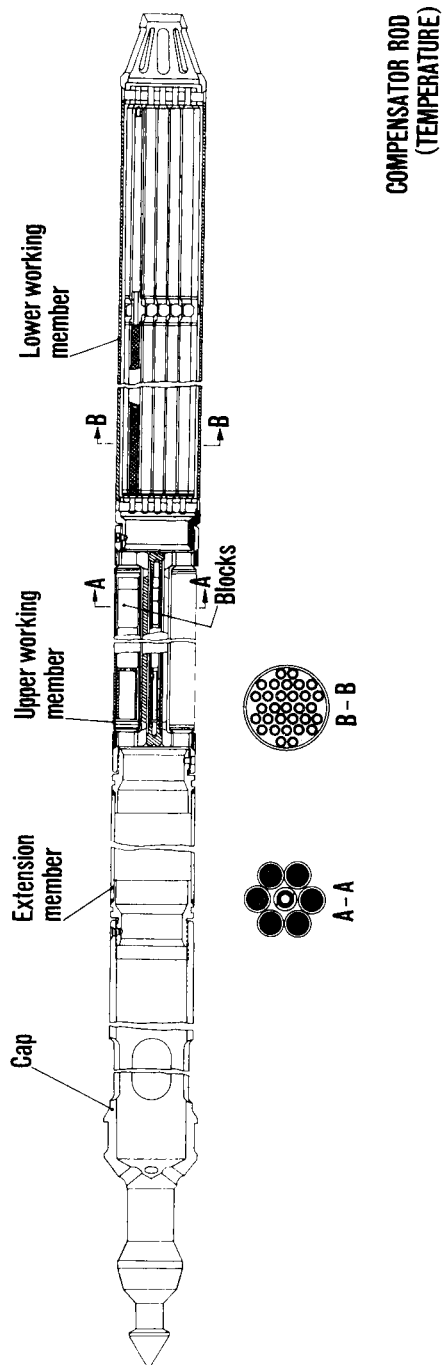


FIG. 7. Compensator rod (temperature).

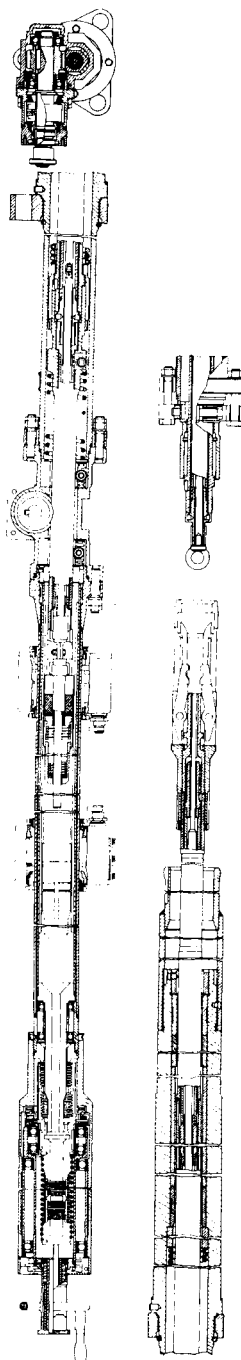


FIG. 8. Rod with rack mechanism for safety member.

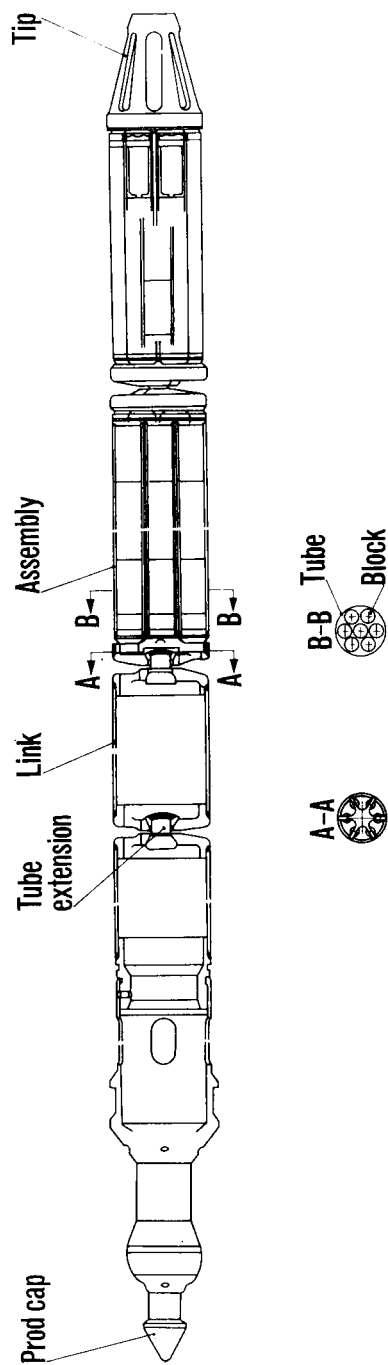


FIG. 9. Safety rod.

temperature field in the reactor, the intermediate heat exchanger, the steam generator and the liquid metal circuits of three-circuit, multi-loop nuclear stations. The dependence of the drop in power in time owing to delayed neutrons and β and γ decays of fission fragments is defined by physics calculations for the reactor. The non-steady state flow is calculated with allowance for the inertia of the flow, forced coasting of the pump and the driving pressure of natural circulation. Heat exchange with the structural elements which are in contact with the sodium is taken into account approximately.

At the same time as the calculation of temperatures, the non-steady state temperature stresses are calculated at decisive points in the circuits. It should be noted that in the plant there are no special systems and removal of the after-heat in any emergency situations is achieved by the main cooling system.

Rapid operation of the emergency system only occurs in exceptional cases. The majority of the signals go to the slow emergency system which reduces the power by a factor of 2.7 in 100 sec. This rate makes it possible to continue shutdown cooling of the reactor with the nominal coolant flow rates without exceeding the permitted thermal stresses in the structural elements.

Total loss of electric power at the main circulating pumps is the most dangerous emergency situation. Therefore, when designing the plant great attention was paid to the problems of ensuring natural circulation. This determined the direction of movement of the coolant in the reactor, the position of the heat-exchange equipment in respect of height, and also affected the design of the intermediate heat-exchanger and the check valve. The flow cross-section of the latter, in the absence of return flow, is never completely covered. The driving pressure of natural circulation ensures a coolant flow of 5–6% of the nominal through the reactor. This is insufficient for shutdown cooling of the reactor initially. Calculations show that with removal of the after-heat solely by means of the inertia of the flow and natural circulation in the individual channels, there can be a rise in the sodium temperature to 800–900°C. Therefore, all available methods of ensuring forced flow of coolant for an initial period of 30–60 sec after rapid shutdown of the reactor were examined; for example, the possibility of using the kinetic energy of the rotating masses of the turbo-generators for extending the operating time of the pumps.

With joint coasting of the pumps and turbo-generators the flow of sodium through the reactor changes very slowly, approximately exponentially with a time constant of 50 sec. Despite the great volume of sodium above the core, with this method of emergency shutdown cooling there is a comparatively abrupt change in the temperature of the coolant at exit from the reactor tank, which causes substantial thermal stresses. Figure 10 shows the variation with time of the non-steady state thermal stresses and coolant temperature at decisive points of the primary circuit. Such conditions of operation are

undesirable and the use of the turbo-generators as emergency drive for the pumps was rejected.

Joint resolution of the equations for the dynamics of the flow and the rotating parts of the pumping unit showed that on disconnection of the power supply the flow of sodium through the reactor changed exponentially

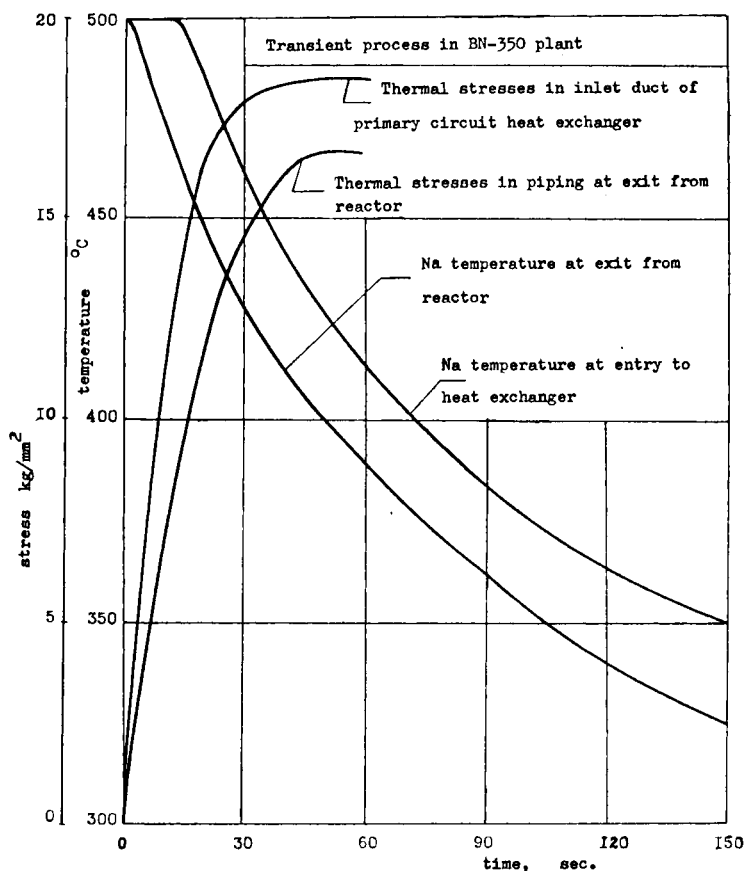


FIG. 10. Coolant temperature and thermal stresses in decisive points of the circuit in shutdown cooling using the energy of turbo-generators coasting.

with a time constant of 7 sec within 15–20 sec. The results of calculations of the temperature conditions in the reactor with removal of the after-heat by coasting of the pumping units and subsequent changeover to natural circulation are given in Fig. 11. The thermal stresses in this case are insignificant and are not shown in the diagram.

In view of the presence of a large reserve of water in the steam generators and the considerable heat capacity of the sodium circuits, the power sources

for supplying the water pumps can be switched on within a few tens of minutes after the fault.

Apart from the emergency conditions, we also studied conditions for the BN-350 reactor in which there is a comparatively small variation in the flow of coolant in the primary circuit. Disconnexion of one primary circuit sodium

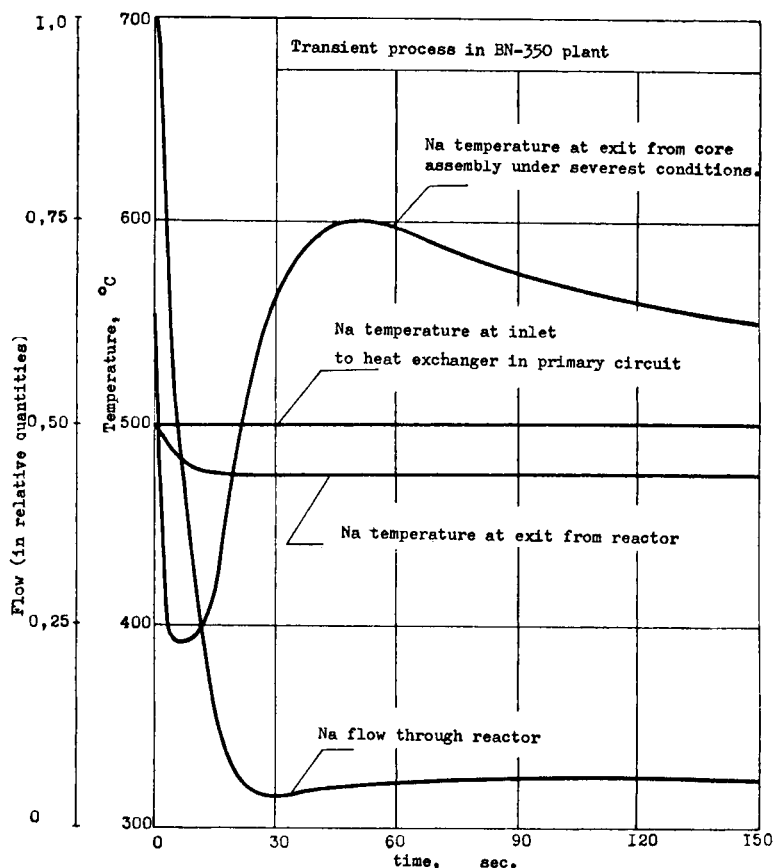


FIG. 11. Variation of the main parameters of the reactor in shutdown cooling by means of pumps coasting and natural circulation.

pump is in practice the most important example of such a change. The main problem in this investigation was to determine the need for shutting down the reactor.

One of the factors which predetermine the necessity of operating the emergency system is the hydrodynamic effects of reactivity, since on variation of the flow of coolant through the reactor it is possible for the fuel assemblies to move, which theoretically can lead to large positive jumps in reactivity,

a considerable rise in the temperature of the fuel elements, and subsequent damage to them.

If, after stoppage of one pump, the reactor is shut down at the former power level, then the difference in the temperatures of all the elements in the reactor will be stabilized within a certain time, reaching values determined by the resultant reduction in flow through the reactor. Such operating conditions are not permissible. Therefore after stoppage of a pump it is necessary to bring the reactor to a reduced power level as soon as possible, giving the nominal level at the new rate of coolant flow. This change can be achieved in different ways. It is possible to introduce an interlock by means of which the power controller will be moved to a new position on the signal that one pump is off. On the other hand, this operation can be carried out by means of the sodium temperature regulator at exit from the reactor. This regulator maintains a given temperature by acting on the power controller and correcting the power so that the temperature does not deviate from the given value. The temperature indicators of the regulator are placed in the reactor tank in the region of the exit ducts. Thus, after a pump is switched off, the increase in the sodium temperature in the reactor tank causes the regulator to operate, and this ultimately shifts the power controller into the position where the sodium temperature is stabilized at the previous level. As a result of the considerable time lags in the reactor tank and in the temperature indicators, the transition to the reduced power will obviously take place more slowly than when an interlock is used.

To find the relationship between the flow of sodium through the reactor and time, the problem of the coasting of one pump connected in parallel to the mains with the others was solved. It was assumed that the hydraulic resistance of all parts of the reactor and piping was proportional to the square of the flow through it, the capacity of the pump varies in proportion with the speed, and the pressure head is proportional to the square of the speed. The kinetic energy of the coolant, pump rotor, and electric motor were taken into account, as was the variation in all the losses in the coasting pump as a function of the speed. The results of calculations of the coasting of the pump are shown in Fig. 12. As a result of the uneven pressures in the clearances, caused by the uneven radial distribution of the flow in the core, by the radial flow of sodium above the tops of the assemblies and other factors, there are movements of the assemblies with respect to the initial position. Since the top ends of the assemblies are free, substantial movements can be expected on variation of the sodium flow through the reactor. Particularly dangerous are the cases when a reduction leads to displacement of the fuel towards the centre of the core.

To determine the movements of the assemblies, the sodium pressure field in the clearances between them was found. The amount of these movements under the action of the forces applied in the radial direction is determined

by the elastic bowing of the assembly and the turning in the support. Ignoring the turning in the support, for the hydrodynamic effects of reactivity corresponding to an increase in the flow of sodium from zero to the nominal flow, the value $\Delta k/k = 3 \times 10^{-4}$ is obtained.

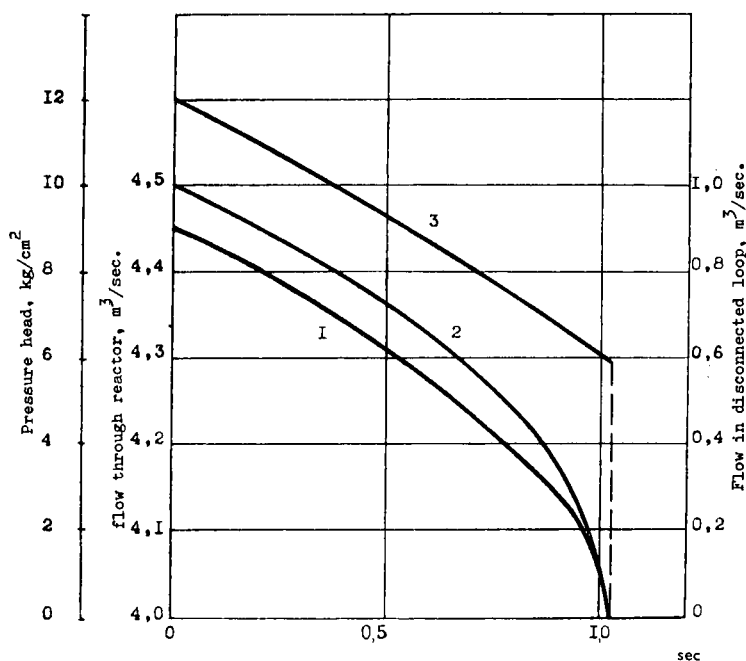


FIG. 12. Coasting of one primary circuit pump.

1. Sodium flow in disconnected loop.
2. Sodium flow through reactor.
3. Pressure head of coasting pump.

The problem of the rigidity of the support of the assemblies has not yet been investigated sufficiently, and it was not possible to take this factor accurately into account at the present time. Therefore in the following calculation an approximation was used. It was assumed that on variation of the flow of sodium through the reactor from zero to the nominal flow, the assemblies were deflected freely in their supports from the vertical position to the extreme outside—as far as the clearances in the supports and between the assemblies would allow. With a corresponding reduction in the flow the assemblies move towards the centre by the same amount. The hydrodynamic effects of reactivity, corresponding to a reduction of the flow from the nominal value to zero with this model were $\Delta k/k = -32 \times 10^{-4}$. The bowing of the

assemblies was not taken into account since it was much less than the assumed movements. From the more accurate calculation it was to be concluded that all the pressure drops in the clearances between the assemblies and, consequently, their movements, were approximately proportional to the square of the sodium flow. Subsequently, therefore, to determine the hydrodynamic effects of reactivity corresponding to a variation in the flow of coolant from the nominal value G_0 to the value G , the following formula was used:

$$\left(\frac{\Delta k}{k}\right)G = -\left(\frac{\Delta k}{k}\right)_0 \cdot \left\{1 - \left(\frac{G^2}{G_0^2}\right)\right\}.$$

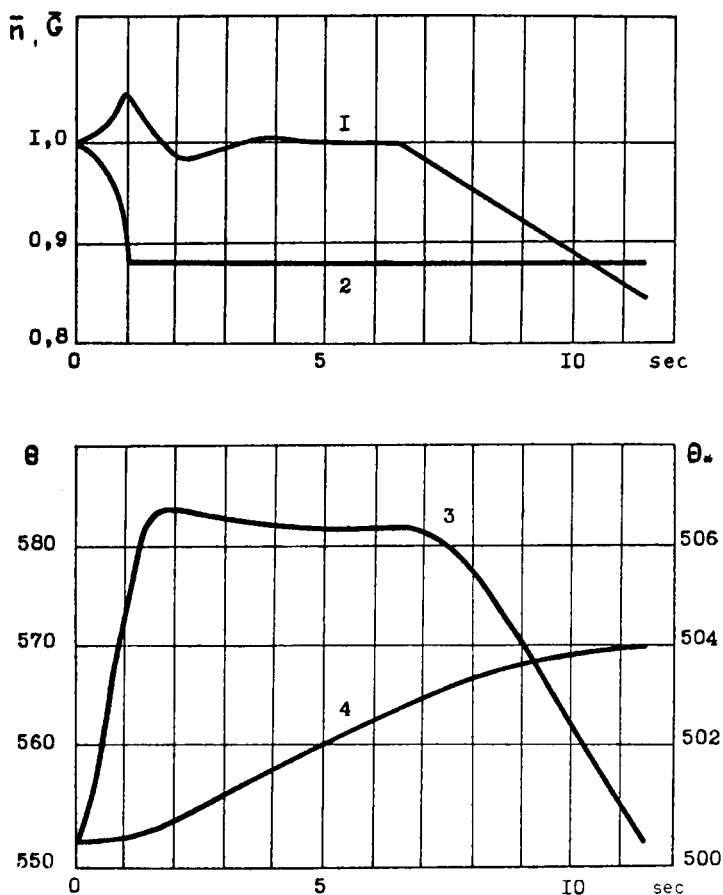


FIG. 13. Transient process in core on stoppage of one primary circuit pump.

1. Power, $\bar{n} = n/n_0$.
2. Coolant flow through reactor, $\bar{G} = G/G_0$.
3. Sodium temperature at exit from core θ .
4. Temperature of indicators in reactor tank, θ^* .

In the BN-350 reactor, on disconnection of one of the five pumps in the primary circuit, the flow of coolant dropped by about 12%. This meant that by the second estimate an introduction of reactivity of 7.2×10^{-4} could be expected.

On the basis of the data obtained on the variation of the sodium flow and the hydrodynamic effects of reactivity, the temperature conditions in the reactor were investigated. The calculations were carried out by a semi-analytical method. The factors associated with the thermal lag of the reactor, mixing chamber, and temperature indicator were taken into account. The results of the calculation are shown in Fig. 13. The calculations showed that the temperature regulator could fully cope with changeover of the plant to a reduced power.

For the case of disconnection of one pump in the primary circuit, the temperature field of the heat exchanger was also investigated. Unless special measures are taken, the sodium temperature at exit from the heat exchanger in the secondary circuit after disconnection of the pump in the primary circuit drops rapidly, which leads to the development of large thermal stresses in the structural elements. The magnitude of these stresses is assessed in the calculation. To reduce them it was found necessary to stop the corresponding pump in the secondary circuit after stoppage of the pump in the primary circuit.

4. Improvement of the Operating Conditions of the BN-350 Fuel Elements

As experience is accumulated on the studies, construction and operation of the reactor, modernization of the reactor installation will be carried out and its economic characteristics improved. Recently the possibilities for improving the operating conditions of the BN-350 fuel elements and increasing the burnup have been examined. For this purpose, with the former dimensions of the assembly unchanged, the possibility of increasing the gas cavity in the fuel element by a corresponding reduction in the thickness of the axial blanket was examined. The top axial reflector is combined with the core in a common enclosure. The calculations showed that a reduction in the thickness of the top reflector to 200 mm, provided there is a slight reduction in the distance between the edge of the core and the edge of axial blanket, does not cause any substantial deterioration in the physical characteristics of the reactor.

The version of the fuel element with an enlarged gas cavity compared with the former version, makes it possible to increase the burnup of the fuel by a factor of 2 and at the same time to reduce the maximum stresses caused by the gas pressure by a factor of about 2.

Combination of the core and the top axial reflector leads to elongation of the closely packed bundle of core cans, and thereby to an undesirable increase

in the pressure drop on the assembly. In view of this, to reduce the stresses induced by pressure, it was proposed to increase the radius of curvature of the angles of the hexagonal tubes. This made it possible to reduce the maximum stresses induced by pressure in the wall of the assembly by a factor of 1.2. The increase in the radius of curvature of the angles of the hexagonal tubes was achieved by removing six (of the 169) corner fuel elements of the assembly. The resultant change in the physical characteristics and dimensions of the core was insignificant.

THE BOR EXPERIMENTAL FAST REACTOR

The BOR reactor—an experimental fast reactor with a maximum power of 60 MW—is being built on the NIIAR site in Melekess. The design and research work was begun at the end of 1963. At the moment the technical design is complete and the working drawings for the main equipment are being prepared. The construction of the building began in May 1965. Start-up of the reactor is planned for 1968.

The BOR reactor is intended for proving and experimentally checking the main decisions in relation to high temperature, high performance, sodium cooled fast reactors for future atomic power stations with an electrical output of 1000 MW and above. The parameters of the first atomic power station with the BN-350 fast reactor were chosen with a certain amount of caution, guaranteeing safe operation of the station, and were based on experiments and operating experience of the BR-5 reactor.

However, a number of measures are already obvious which must be taken in the first place to improve the physical and technical/economic characteristics of atomic power stations with fast reactors. These measures include:

1. Increasing the burnup.
2. Increasing the power density.
3. Increasing the coolant temperature at exit from the reactor and thereby raising the efficiency of the steam turbine cycle.
4. Increasing the unit power of the station.

The most important item is the creation and development of fuel elements for the cores of high performance reactors, fuel elements which will operate reliably at high thermal flux densities and high temperatures. The BOR reactor is being built in the first place to solve this problem. Fuel elements of different composition and design will be used in it, including vented ones.

1. Core and Blankets

The core of the BOR reactor will consist of hexagonal assemblies with a dimension across the flats of 44 mm, arranged in a pitch of 45 mm. The number of assemblies will vary according to the operating conditions, the

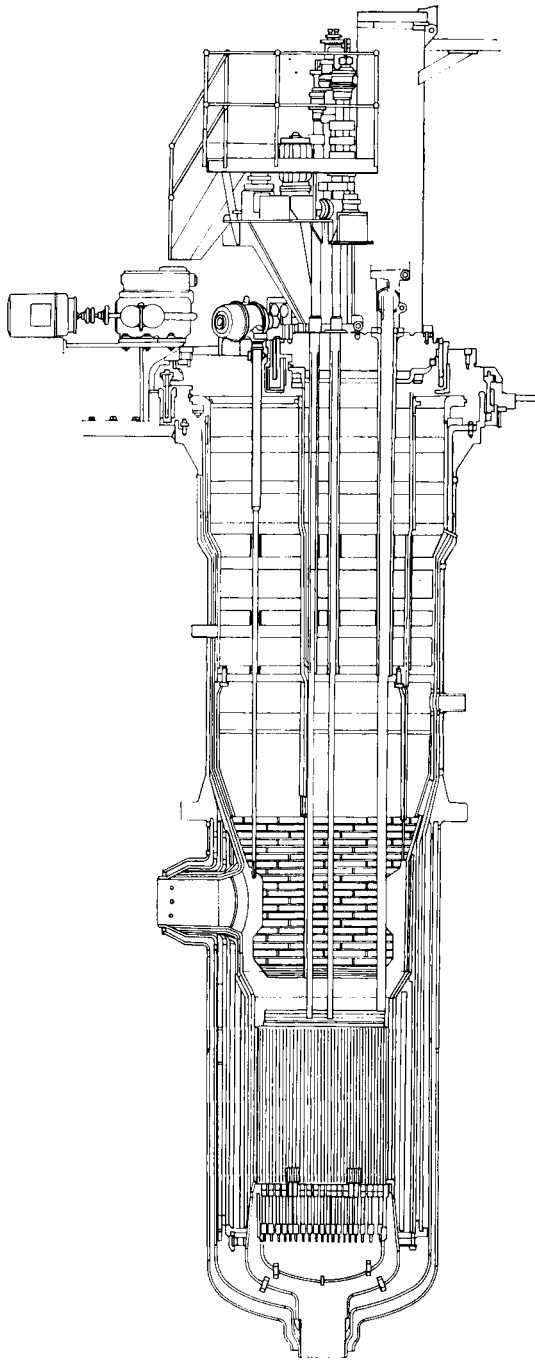


FIG. 14. General view of BOR reactor.

burnup, and the type of fuel. The version with new 90% enriched uranium dioxide fuel amounts to seventy to eighty assemblies, which corresponds to a 150–160 kg charge of U-235. In the working section of the assembly there are thirty-seven fuel elements 6.1 mm in diameter arranged in a triangular lattice with a pitch of 6.7 mm. The fissile length of the fuel element is 400 mm. As fissile material it is possible to use enriched uranium or plutonium oxide, uranium or plutonium carbide and other compositions. The cans used for the fuel elements are stainless steel tubes with a wall thickness of 0.4 mm for the sealed version and 0.25 mm for the vented version. In the sealed version there are cavities in the fuel elements for collecting the gaseous fission products; in the vented version there are special arrangements for releasing them into the coolant. The version of the assembly with vented fuel elements is shown in Fig. 15. The elements are fastened at the bottom to grids welded to the hexagonal tube. The top ends are free and can undergo axial movements. Spacing of the fuel elements is achieved by means of wires wound in spirals on the can. Above and below the active part of the assembly are the axial blankets. Their design depends on the design of the fuel elements of the active part of the assembly.

The fuel elements are enclosed in a hexagonal can 1 mm thick. The assembly has a top with a cap for gripping by the refuelling machine and bottom end fitting for setting the assembly in the throttle sleeves of the pressure header. The bottom end fitting has side holes for passage of sodium, which prevents the possibility of the assembly lifting under the effect of the flow of coolant.

The top ends of the assemblies are not fastened, which ensures the possibility of free thermal elongation in the axial direction; movement in the radial direction is possible only within the limits of the engineering tolerances. Throttling of the flow of sodium along the assemblies in accordance with their heat output is achieved by means of slots in the throttling sleeves of the pressure header, covering a given number of side holes in the bottom ends of the assemblies.

The radial blanket is made up of assemblies with an outside configuration similar to the core assemblies. Each of them contains seven elements 14.5 mm in diameter arranged with a pitch of 15.2 mm in a triangular lattice. The can thickness of the element is 0.35 mm. The elements contain pellets of depleted uranium dioxide and are 900 mm long with respect to the dioxide. The mean blanket thickness is 150 mm. The assemblies are fitted with their bottom end fittings in the throttling sleeves of the pressure header, and the top ends are free as in the core assemblies. Sodium is supplied to the assembly from a low pressure header via a throttling disk in the bottom end fitting. Throttling of the flow of coolant through the assemblies of the radial blanket is achieved by varying the diameter of this disk. There are 259 assemblies altogether in the core and blanket.

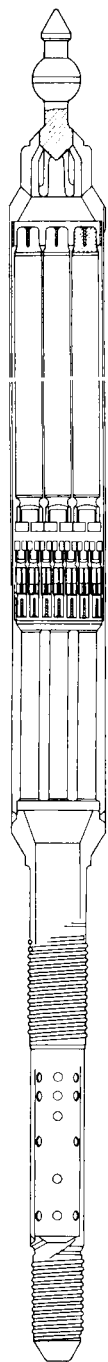


Fig. 15. BOR reactor core assembly.

Vessel and Internal Equipment (Fig. 15)

The reactor vessel is cylindrical. The diameter at core level is 1100 mm and the maximum diameter is 1500 mm. The vessel is 6500 mm high, and wall thickness is 16–20 mm. The central and bottom part are enclosed in a leak-tight housing preventing the reactor being drained in the event of a leak. The housing is also used for warming up the reactor with gas.

The vessel has one inlet duct 300 mm in diameter, welded into the elliptical end, and two outlet ducts also 300 mm in diameter just above the level of the tops of the assemblies.

At the top the reactor is covered by two eccentrically placed rotating plugs for bringing the refuelling devices to the given cell of the core or blanket. The plugs are made of alternate layers of steel and borated graphite and provide the necessary biological shielding. Gas-tight sealing of the plugs is achieved by the use of freeze seals filled with a bismuth–tin alloy; during operation of the reactor this alloy is in the solidified condition. On the small rotating plug there are all the drives of the safety and control elements and the transfer box for connecting the reactor to the refuelling machine.

Inside the reactor vessel are the pressure header with the core and blankets, the basket, and thermal shield. The pressure header is divided into a high pressure zone, from which the core assemblies are supplied, and a low pressure zone for the blanket elements. The header is suspended on the basket, which is a stepped cylinder fastened to the top of the vessel. Where necessary they can be replaced together with the header. A small part of the main sodium flow passes between the basket and the vessel to cool the latter.

At the bottom and central parts of the basket are the annular steel reflectors of the thermal and neutron shielding with a total thickness of 240 mm. They are used to reduce the neutron flux impinging on the reactor vessel.

2. Control and Safety System

The control and safety system of the reactor includes three independent safety systems, two compensators, one of which is a temperature compensator, and two automatic power controllers. The safety and compensation rods are in the core and the control rods in the radial blanket. The operating elements of all systems are identical in design and consist of a bundle of boron-containing rods. Each rod has a stainless steel can with a diameter of 12×0.4 mm filled with boron carbide enriched in B-10. The effectiveness of the safety and control elements and the value of the different reactivity effects are given in Table 4. The operating elements are connected via a rod with grabs to servo-drives on the rotating plug. When it is necessary to rotate the plugs, the elements are lowered into the lowest position and are disengaged from their drives.

TABLE 4. MAIN DESIGN CHARACTERISTICS OF THE BOR REACTOR

Maximum thermal power	60 MW
Sodium temperature at entry	360–450°C
Mean temperature at exit	600°C
Flow through reactor	750–1000 t/h
Core diameter (middle)	41 cm
Core height	40 cm
Mean specific power	900 kW/l
Maximum heat flux	2.65×10^6 Kcal/m ² h
Maximum sodium speed	10 m/s
Maximum can temperature with allowance for possible overheating	800°C
Effectiveness of three safety rods	3.3%
Effectiveness of temperature compensator	1.0%
Effectiveness of burnup compensator	1.7%
Effectiveness of two control rods	0.48%
Temperature coefficient of reactivity	-4×10^{-5} l/°C
Power coefficient of reactivity	-1.5×10^{-4} l/MW

3. Refuelling System

The charging of new assemblies and the discharging of the spent ones is carried out via a special channel in the small rotating plug. By rotating the large and small plugs, the channel is guided to the chosen socket in the core or blanket. Then the charge–discharge machine is moved up to it. This is a container with gripping devices, mounted on a gantry crane and able to move along co-ordinate axes. The charge–discharge machine is connected and sealed to the channel in the small plug; by means of the grab the assembly to be replaced is raised into the shielded container with an inert atmosphere, in which it is transferred to an external store.

To take away the after-heat there is a closed gas circuit provided for in the design of the charge–discharge machine, with the corresponding equipment. Transportation and insertion of new assemblies into the reactor are carried out by the same machine. The assemblies are previously heated to about 300°C in an electric furnace.

4. Cooling System and Equipment

A three-circuit system was adopted for the BOR reactor: sodium, sodium, water–steam. Removal of the heat is carried out in two loops, each of which is designed for 50% of the maximum power. The primary circuit loop includes a centrifugal submerged pump, intermediate heat exchanger, check valve and shut-off fittings.

In each secondary circuit loop, apart from the heat exchanger and pump there is a steam generator. In addition there is an air cooled heat exchanger

designed for a thermal capacity of 30 MW, which can be connected to either secondary circuit loop instead of the corresponding steam generator. The flow of air through the heat exchanger is produced by fans. The steam produced in the steam generators, with parameters of 540°C and 100 atm is used in a standard turbo-generator to produce electrical energy.

The arrangement of the first, second and third circuits is such that the necessary natural circulation flows are produced, guaranteeing reliable long term heat removal from the reactor.

Intermediate Heat Exchanger (Fig. 16)

The intermediate heat exchanger is a vertical cylindrical vessel in which there is a tube bundle with floating top. The primary circuit sodium flows in the space between the tubes from top to bottom, the secondary sodium flows up the tubes. The heat exchanger is mounted on the top cover of the primary circuit boxes and is designed so that the tube bundle with the cover and built in top shield can be removed without cutting the primary circuit piping or entering buildings with a high level of radiation.

Steam Generators

The steam generators fitted in the different loops of the BOR reactor are of different designs. In one loop there will be a steam generator with natural circulation, in another there will be a once-through type. This is done so that different versions can be developed in conditions as close as possible to the operating conditions of future power systems. A feature of both types of steam generator is the absence of free sodium levels in their vessels, and the removal of these levels to special buffer tanks installed at the sodium exit. The possibility of using this system is now being checked experimentally.

Steam Generator with Natural Circulation (Fig. 17)

The steam generator with natural circulation has an evaporator with straight tubes and a U-shaped steam superheater. The water and steam in both cases flow in the tubes, and the sodium flows in the space between. Water from the drum is fed to the lower distribution chamber of the evaporator and the steam/water mixture is led off from the top chamber to a drum. The sodium in the evaporator flows from top to bottom. To compensate the different temperature elongations of the tubes and vessel, lenticular compensators are provided on the latter. To prevent moisture entering the steam superheater, there are appropriate separators inside the drum. The material for the evaporator and for the steam superheat is perlitic steel.

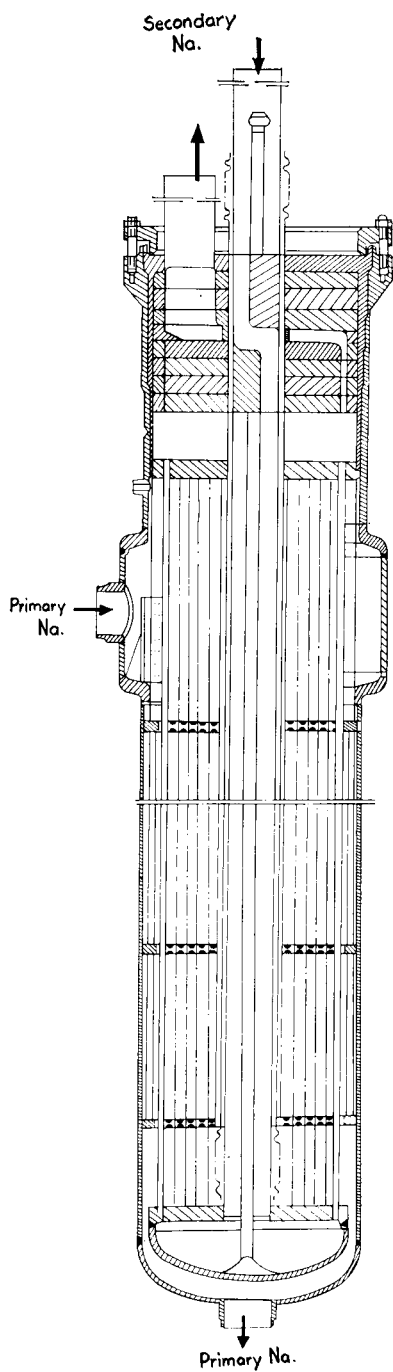


FIG. 16. Intermediate heat exchanger.

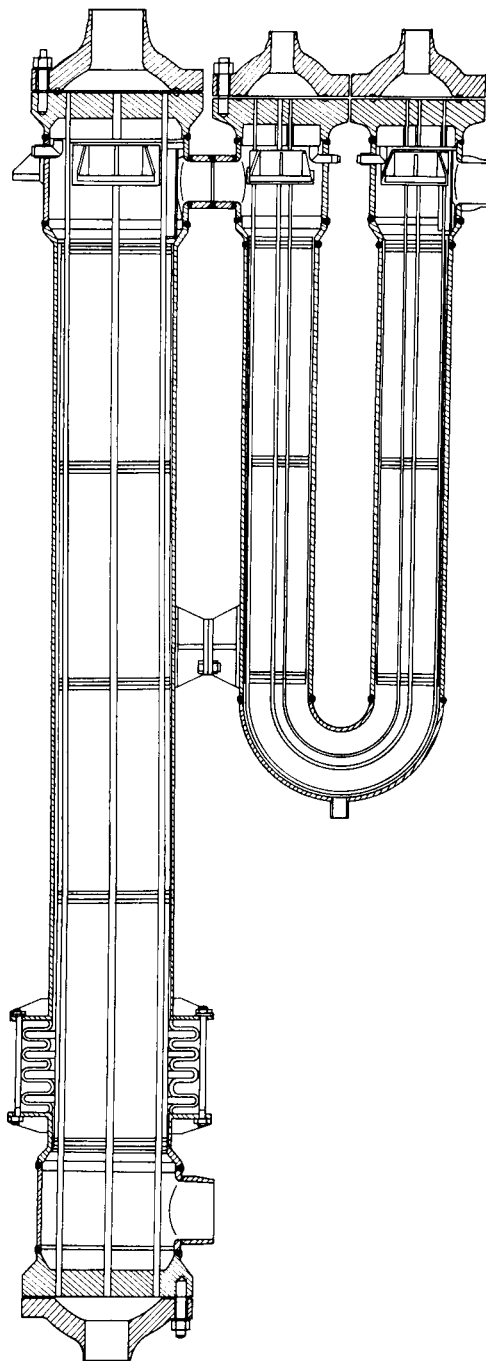


FIG. 17. Steam generator with natural circulation in the BOR installation.

Once-through Steam Generator (Fig. 18)

The once-through steam generator consists of two cylindrical vessels about 1 m in diameter. The tube bundle of the evaporator consists of sixty coils, and in the steam superheater it consists of ninety coils made of 16×2.5 mm tubing. The tube plates in each vessel are fitted with one at the top and one at the bottom. The direction of the sodium flow in the steam generator is from bottom to top, and in the evaporator from top to bottom; the water and steam move in the opposite direction. This system of movement of the sodium and water in the evaporator makes it possible to arrange natural circulation in it in some emergency conditions. For this purpose there is a special small-sized compensator cooled by water from a permanent power supply source. The steam generator is made of perlitic steel.

Circulating Pumps

The circulating pumps of the primary and secondary circuits are similar in design. Each pump has two bearings—a bottom radial hydrostatic bearing operating in the sodium, and a top journal thrust oil bearing, and a gas-tight oil seal on the shaft. The metal enters the intake directly from the pump tank, and the necessary flow through the intake lines is produced by the difference in levels in the pump tank and reactor. The speed is controlled by using d.c. motors. The pump is fitted on the upper cover of the primary circuit boxes and the motor is accessible for servicing during operation.

5. Experimental Facilities

On the BOR reactor there are great opportunities for carrying out experimental and research work on neutron physics, the behaviour of fuel at high burnups, radiation resistance of materials, the study of the escape of fission products into the sodium coolant and gas system of the reactor and their distribution through the circuit. The latter problems are of particular interest, since vented fuel elements will be tested and developed in the BOR. For studying a whole set of problems on the reactor, by-pass sections will be fitted with spectrometer indicators for monitoring the coolant contamination level and the circuit surface during operation of the reactor; gas lines for spectrometric analysis of fission products in the gas cavities of the reactor and pumps; delayed neutron detectors are installed in the exit piping. For continuous monitoring of the level of the gaseous activity, including short-lived products, there is provision for circulation of gas along gas lines.

Another facility of great interest is the special channel near the centre of the core for studying the operation of fuel elements. The channel is an

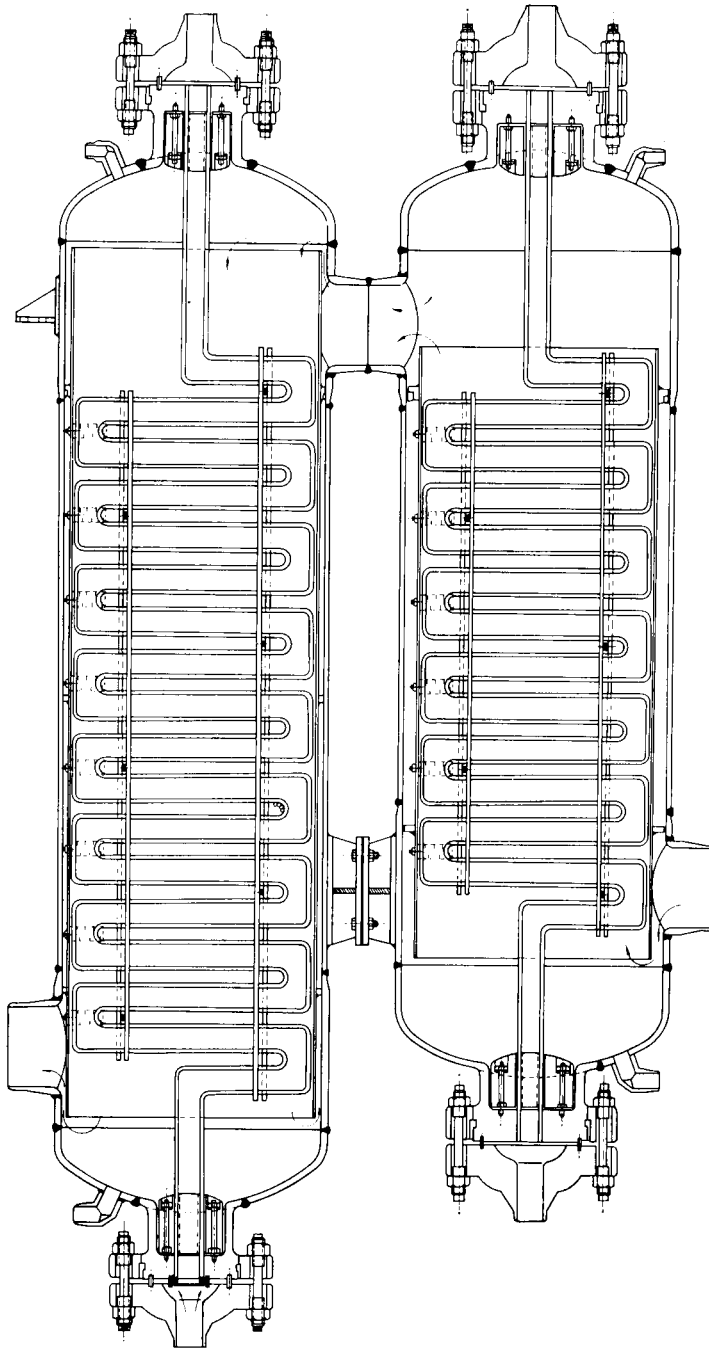


FIG. 18. Steam generator of BOR once-through installation.

assembly identical with the rest in external outline but fitted with the appropriate indicators and fastened to a rod through which the leads to these indicators are brought out. During discharging the assembly is removed from the core and rotates with the plugs.

In addition there are four horizontal bundles for physical investigations, and ten vertical uncooled channels, including those with a diameter of 300 mm. Into the core and blanket it is possible to load a large number of assemblies with specimens—up to thirty assemblies at the same time in the core.